

$$\begin{bmatrix} \cos 90^\circ & \sin 90^\circ \\ -\sin 90^\circ & \cos 90^\circ \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

<https://xkcd.com/184/>

Classic Mechs EP3: Eigenvector and Higher Order Linear ODE Systems

Classic Mechs EP3.1: higher order ODE systems

Learning objectives

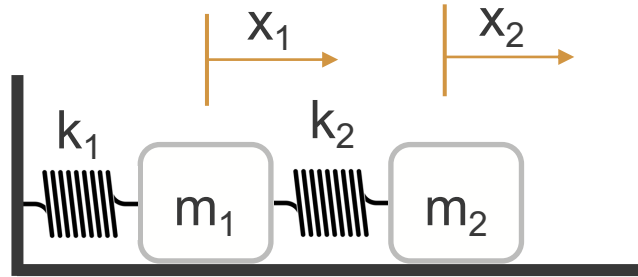
- 1) Work on double harmonic oscillators and derive the slightly more messy 4th order ODE.
- 2) Back to the 1st order land: derive the general state space matrix form for higher order linear ODE systems.
- 3) Motivation to use the eigenvalue and eigenvector method to solve such systems of equations.

What's next for next week?

Let's start to move into the Lagrange land, and revisit our old friend of rotational motion, energy, conservative force... along the way.

Double harmonic oscillators = 4th order

Double harmonic oscillators case A:



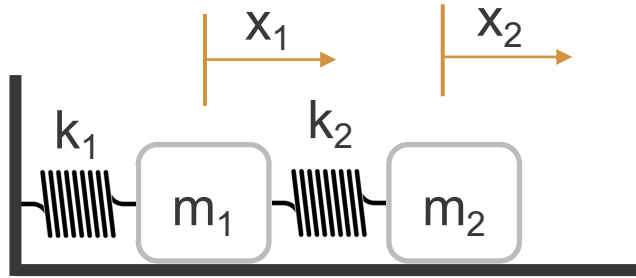
$$m_1 \ddot{x}_1 = -k_1 x_1 + k_2 (x_2 - x_1)$$

$$m_2 \ddot{x}_2 = -k_2 (x_2 - x_1)$$



Double harmonic oscillators = 4th order

Double harmonic oscillators case A:



$$m_1 \ddot{x}_1 = -k_1 x_1 + k_2 (x_2 - x_1)$$

$$m_2 \ddot{x}_2 = -k_2 (x_2 - x_1)$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{v}_1 \\ \dot{x}_2 \\ \dot{v}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -\frac{k_1+k_2}{m_1} & 0 & \frac{k_2}{m_1} & 0 \\ 0 & 0 & 0 & 1 \\ \frac{k_2}{m_2} & 0 & -\frac{k_2}{m_2} & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ v_1 \\ x_2 \\ v_2 \end{bmatrix}$$

1st order system of equations not so bad now, heh?

Use systems of equations to tackle higher order systems

$a_n x^{(n)} + a_{n-1} x^{(n-1)} + \dots + a_2 \ddot{x} + a_1 \dot{x} + a_0 x = 0$ General form of higher order linear ODE

$a_n \lambda^n + a_{n-1} \lambda^{n-1} + \dots + a_2 \lambda^2 + a_1 \lambda + a_0 = 0$ Characteristic equation

$X(t) = C_1 e^{\lambda_1 t} + C_2 e^{\lambda_2 t} + \dots + C_n e^{\lambda_n t}$ General form of solution

How many initial conditions we need to determine the constants of $C_1, C_2, \dots, C_n$?

How to write the higher order ODE in matrix system from? $\underline{\dot{x}} = \underline{\underline{A}} \underline{x}$

$$a_n x^{(n)} + a_{n-1} x^{(n-1)} + \dots + a_2 \ddot{x} + a_1 \dot{x} + a_0 x = 0$$

It's time to solve general systems of equations: $\underline{\dot{x}} = \underline{A} \underline{x}$

Case 1: uncoupled systems

It's time to solve general systems of equations: $\underline{\dot{x}} = \underline{A} \underline{x}$

Case 2: coupled systems